

Nonparametric estimators for quantiles of the conditional residual lifetime distribution

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Introduction

- ▶ In medical research, one is often interested in studying **association between event times T_1 and T_2** , potentially subject to (right-)censoring
- ▶ Strength of association can be expressed in terms of **global and local association measures**, e.g.,
 - ▶ Kendall's tau
 - ▶ Cross-ratio function (Clayton, 1978)
 - ▶ etc.
- ▶ Alternatively, one can study **(quantiles of) the conditional residual lifetime distribution** (e.g., Abrams et al., 2021):

Example

What is the median residual time to the occurrence of an event (i.e., represented by T_1), given that $T_2 \in (t_{21}, t_{22}]$

Notation:

- ▶ Let (T_1, T_2) represent a random vector with $T_1 \geq 0$, $T_2 \geq 0$ and
 - ▶ $F(t_1, t_2) = P(T_1 \leq t_1, T_2 \leq t_2)$ the **joint cumulative distribution function**;
 - ▶ $F_1(t_1) = P(T_1 \leq t_1)$ and $F_2(t_2) = P(T_2 \leq t_2)$ **marginal distribution functions**
- ▶ Conditional distribution of T_1 given that $T_2 \leq t_2$:

$$F(t_1 | T_2 \leq t_2) = P(T_1 \leq t_1 | T_2 \leq t_2) = \frac{F(t_1, t_2)}{F_2(t_2)} =: \tilde{F}_{t_2}(t_1)$$

- ▶ **Conditional residual lifetime distribution** of T_1 at t_1 , given that $T_2 \leq t_2$:

$$P(T_1 - t_1 \leq y | T_1 > t_1, T_2 \leq t_2) = \frac{\tilde{F}_{t_2}(t_1 + y) - \tilde{F}_{t_2}(t_1)}{1 - \tilde{F}_{t_2}(t_1)}$$

- For $0 < p < 1$, we define the p -th quantile of the conditional residual lifetime of T_1 , given that $T_2 \leq t_2$ as:

$$\begin{aligned} Q(p | t_1, t_2) &= \inf \left\{ y : \frac{\tilde{F}_{t_2}(t_1 + y) - \tilde{F}_{t_2}(t_1)}{1 - \tilde{F}_{t_2}(t_1)} \geq p \right\} \\ &= -t_1 + \tilde{F}_{t_2}^{-1} \left[p + (1 - p)\tilde{F}_{t_2}(t_1) \right] \end{aligned}$$

- Nonparametric estimation of $Q(p | t_1, t_2)$ requires **nonparametric estimators for the conditional distribution function** $\tilde{F}_{t_2}$ and its quantile function (relying on the Akritas-Van Keilegom estimator for $F(t_1, t_2)$; see, e.g., Abrams et al., 2021)
- Abrams et al. (2021): estimators under **one-component censoring in T_1** only; asymptotic normality of the estimator

- ▶ An **extension of the work by Abrams et al. (2021)** to account for
 - ▶ Univariate censoring next to one-component censoring
 - ▶ Conditioning on T_2 of the form $\{t_{21} < T_2 \leq t_{22}\}$ instead of $\{T_2 \leq t_2\}$ (or $T_2 > t_2$)
 - ▶ Direct estimators for the joint distribution function $F(t_1, t_2)$ instead of the Akritas-Van Keilegom estimator
- ▶ We propose:
 - ▶ Two estimators for bivariate time-to-event data subject to **one-component censoring**
 - ▶ Two estimators for bivariate time-to-event data subject to **univariate censoring**

Motivating example

- ▶ Data from a **randomized controlled trial** evaluating the difference between the **main treatment (chemotherapy)** and **adjuvant treatment (chemotherapy and hormone treatment with tamoxifen)** for node-positive breast cancer patients (Schumacher et al., 1994)
- ▶ The event time of interest T_1 is the **time to tumor recurrence (defined as the occurrence of locoregional or distant recurrence, contralateral tumor or a secondary tumor)**
- ▶ T_2 is defined as a **risk score based on potentially prognostic factors** that were previously assessed and identified in Sauerbrei et al. (1999)

- In total, **data on 686 patients** are available of whom 299 (43.6%) suffered from tumor recurrence and 246 received hormone therapy

Recurrence	Hormonal therapy		Total
	No	Yes	
No	235	152	387
Yes	205	94	299
Total	440	246	686

- **Goal:** Estimate quantiles of the conditional residual time to tumor recurrence for different risk groups (based on T_2)

Conditional residual lifetime distribution

- Conditional distribution of T_1 given that $t_{21} < T_2 \leq t_{22}$:

$$\begin{aligned}\tilde{F}(t_1 \mid t_{21}, t_{22}) &:= P(T_1 \leq t_1 \mid t_{21} < T_2 \leq t_{22}) \\ &= \frac{F(t_1, t_{22}) - F(t_1, t_{21})}{F_2(t_{22}) - F_2(t_{21})}\end{aligned}\tag{1}$$

- **Conditional residual lifetime distribution** of T_1 at t_1 , given that $t_{21} < T_2 \leq t_{22}$:

$$\begin{aligned}F(y \mid t_1, t_{21}, t_{22}) &= P(T_1 - t_1 \leq y \mid T_1 > t_1, t_{21} < T_2 \leq t_{22}) \\ &= \frac{\tilde{F}(t_1 + y \mid t_{21}, t_{22}) - \tilde{F}(t_1 \mid t_{21}, t_{22})}{1 - \tilde{F}(t_1 \mid t_{21}, t_{22})}\end{aligned}$$

- For $0 < p < 1$, we define the p -th quantile of the conditional residual lifetime of T_1 , given that $t_{21} < T_2 \leq t_{22}$ as:

$$\begin{aligned} Q(p \mid t_1, t_{21}, t_{22}) &= \inf \{y : F(y \mid t_1, t_{21}, t_{22}) \geq p\} \\ &= -t_1 + \tilde{F}^{-1} \left[p + (1 - p) \tilde{F}(t_1 \mid t_{21}, t_{22}) \mid t_{21}, t_{22} \right], \end{aligned} \quad (2)$$

with $\tilde{F}^{-1}$ the inverse of $\tilde{F}$.

Estimators of the quantile function

- ▶ Our estimators for $F(t_1, t_2)$ under **one-component and univariate censoring** will follow as particular cases of two more general estimators for $F(t_1, t_2)$
- ▶ (T_1, T_2) is subject to **independent censoring** by a vector (C_1, C_2)
- ▶ The **observable variables** are $(Z_1, Z_2, \Delta_1, \Delta_2)$ where
 - ▶ Observation times $Z_1 = T_1 \wedge C_1$ and $Z_2 = T_2 \wedge C_2$
 - ▶ Censoring indicators $\Delta_1 = I(T_1 \leq C_1)$ and $\Delta_2 = I(T_2 \leq C_2)$

① Inverse-probability weighted (IPW) estimator:

- ▶ The estimator of Burke (1988) is derived from the theoretical relation

$$F(t_1, t_2) = \int_0^{t_1} \int_0^{t_2} \frac{dP(T_1 \leq y_1, T_2 \leq y_2, \Delta_1 = 1, \Delta_2 = 1)}{S_G(y_1-, y_2-)}$$

- ▶ The **corresponding estimator** for a sample $(Z_{1i}, Z_{2i}, \Delta_{1i}, \Delta_{2i})$, $i = 1, \dots, n$, is

$$\hat{F}_B(t_1, t_2) = \frac{1}{n} \sum_{i=1}^n \frac{\Delta_{1i} \Delta_{2i}}{\hat{S}_G(Z_{1i-}, Z_{2i-})} I(Z_{1i} \leq t_1, Z_{2i} \leq t_2),$$

where $\hat{S}_G(t_1, t_2)$ is a nonparametric estimator for $S_G(t_1, t_2) = P(C_1 > t_1, C_2 > t_2)$

① Inverse-probability weighted (IPW) estimator:

- Under **one-component censoring** with $Z_2 = T_2$ and $\Delta_2 = 1$, we have

$$\hat{F}_{B1}(t_1, t_2) = \frac{1}{n} \sum_{i=1}^n \frac{\Delta_{1i}}{1 - \hat{G}(Z_{1i}-)} I(Z_{1i} \leq t_1, T_{2i} \leq t_2),$$

where $\hat{G}$ the Kaplan-Meier estimator for $G(c) = P(C_1 \leq c)$ based on the observations $(Z_{1i}, 1 - \Delta_{1i})$ ($i = 1, \dots, n$)

- Under **univariate censoring** with $C_1 = C_2 = C$ and $S_G(t_1, t_2) = 1 - G[(t_1 \vee t_2)-]$, we have

$$\hat{F}_{B2}(t_1, t_2) = \frac{1}{n} \sum_{i=1}^n \frac{\Delta_{1i} \Delta_{2i}}{1 - \hat{G}[(Z_{1i} \vee Z_{2i})-]} I(Z_{1i} \leq t_1, Z_{2i} \leq t_2),$$

where $\hat{G}$ is the Kaplan-Meier estimator of G based on the observations $((T_{1i} \vee T_{2i}) \wedge C_i, \Delta_i = 1 - \Delta_{1i} \Delta_{2i}) = (Z_{1i} \vee Z_{2i}, \Delta_i)$ ($i = 1, \dots, n$)

② Lin-Ying estimator:

- The estimator of Lin and Ying is derived from the following theoretical relation

$$P(Z_1 > t_1, Z_2 > t_2) = S(t_1, t_2)S_G(t_1, t_2),$$

where $S(t_1, t_2) = P(T_1 > t_1, T_2 > t_2)$ is the joint survival function of (T_1, T_2)

- Using this relation, $S(t_1, t_2)$ is estimated as

$$\hat{S}_{LY}(t_1, t_2) = \frac{1}{n} \frac{1}{\hat{S}_G(t_1, t_2)} \sum_{i=1}^n I(Z_{1i} > t_1, Z_{2i} > t_2),$$

where $\hat{S}_G(t_1, t_2)$ is a suitable nonparametric estimator of $S_G(t_1, t_2)$.

- This leads to an estimator for $F(t_1, t_2)$ using

$$\hat{F}_{LY}(t_1, t_2) = \hat{S}_{LY}(t_1, t_2) - \hat{S}_{LY}(t_1, 0) - \hat{S}_{LY}(0, t_2) + 1$$

Estimators of the joint distribution function $F(t_1, t_2)$

② Lin-Ying estimator:

- For **one-component censoring**, we have:

$$\begin{aligned}\hat{F}_{LY1}(t_1, t_2) = & \left\{ \frac{1}{n} \sum_{i=1}^n I(T_{2i} \leq t_2) \right\} \\ & - \left\{ \frac{1}{n} \frac{1}{[1 - \hat{G}(t_1)]} \sum_{i=1}^n I(Z_{1i} > t_1, T_{2i} \leq t_2) \right\}\end{aligned}$$

- For **univariate censoring**, we have:

$$\begin{aligned}\hat{F}_{LY2}(t_1, t_2) = & \left\{ \frac{1}{n} \frac{1}{[1 - \hat{G}(t_1 \vee t_2)]} \sum_{i=1}^n I(Z_{1i} > t_1, Z_{2i} > t_2) \right\} \\ & - \left\{ \frac{1}{n} \frac{1}{[1 - \hat{G}(t_1)]} \sum_{i=1}^n I(Z_{1i} > t_1) \right\} \\ & - \left\{ \frac{1}{n} \frac{1}{[1 - \hat{G}(t_2)]} \sum_{i=1}^n I(Z_{2i} > t_2) \right\} + 1\end{aligned}$$

- ▶ For the **numerator in equation (1)** we use $\hat{F}(t_1, t_{22}) - \hat{F}(t_1, t_{21})$
- ▶ The **denomintator in equation (1)** is estimated by the difference of the Kaplan-Meier estimators for $F_2(t_2)$, evaluated at t_{22} and t_{21} , respectively
- ▶ Combining the four estimators for the numerator with the appropriate estimator for the denominator leads to estimators $\tilde{F}_n$ for $\tilde{F}$.
- ▶ Finally, plugging in each of these estimators $\tilde{F}_n$ (and $\tilde{F}_n^{-1}$ in equation (2)), leads to four estimators $Q_n(p \mid t_1, t_{21}, t_{22})$ for the quantile function $Q(p \mid t_1, t_{21}, t_{22})$:

$$Q_n(p \mid t_1, t_{21}, t_{22}) = -t_1 + \tilde{F}_n^{-1}(p + (1 - p)\tilde{F}_n(t_1 \mid t_{21}, t_{22}) \mid t_{21}, t_{22}).$$

Simulations

- For T_j , $j = 1, 2$, we use a **Weibull distribution** with $s_j = 1.5$ and $d_j = 0.5$ such that

$$F_j(t_j) = 1 - \exp(-d_j t_j^{s_j}), \quad j = 1, 2$$

- For $C = C_1$ (one-component censoring) or $C = C_1 = C_2$ (univariate censoring) we use a Weibull distribution with parameters $s_C = 1.5$ and $d_C = 0.15$ (**moderate censoring**)
- To arrive at **bivariate Weibull distributions**, we use the **Clayton copula**

$$C_\theta(u, v) = (u^{-\theta} + v^{-\theta} - 1)^{-1/\theta}, \quad \theta \geq 0,$$

such that $F(t_1, t_2) = C_\theta[F_1(t_1), F_2(t_2)]$

- We generate **$M = 500$ datasets** of **$n = 250$ observations** $(z_{1i}, z_{2i}, \delta_{1i}, \delta_{2i})$, $i = 1, \dots, n$ using the Clayton copula with θ corresponding to Kendall's τ equal to 0.2 ($\theta = 0.5$)
- **Other scenarios** (heavy censoring, different copulae, etc.) not shown

Simulation results – One-component censoring

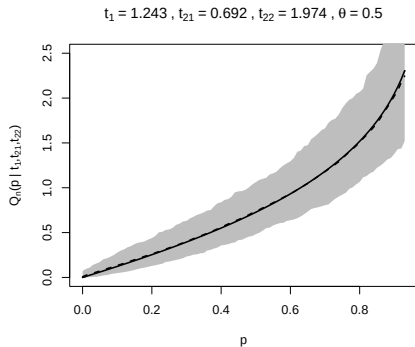
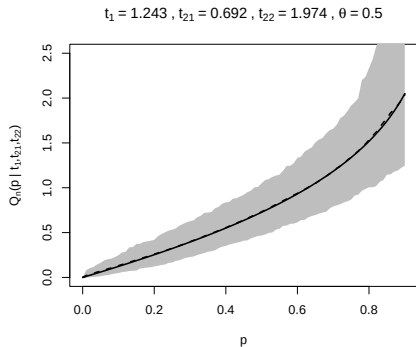


Figure: True quantile function $Q(p | t_1, t_{21}, t_{22})$ (black solid line) and averaged estimated quantile function $Q_n(p | t_1, t_{21}, t_{22})$ as a function of p (black dashed lines) for $t_1 = d_1^{-1/s_1} [\ln(2)]^{1/s_1} \approx 1.243$, $t_{21} \approx 0.692$ and $t_{22} \approx 1.974$ (moderate one-component censoring).

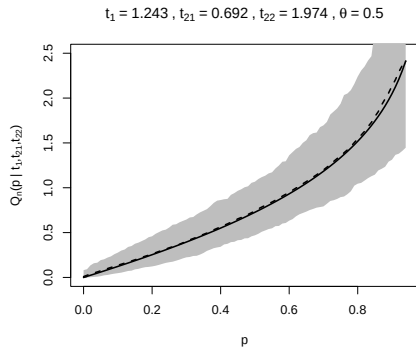
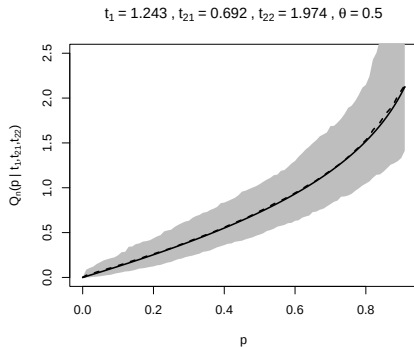


Figure: True quantile function $Q(p | t_1, t_{21}, t_{22})$ (black solid line) and averaged estimated quantile function $Q_n(p | t_1, t_{21}, t_{22})$ as a function of p (black dashed lines) for $t_1 = d_1^{-1/s_1} [\ln(2)]^{1/s_1} \approx 1.243$, $t_{21} \approx 0.692$ and $t_{22} \approx 1.974$ (moderate univariate censoring).

Data application

- ▶ We consider a **generalized additive Cox proportional hazards model** including non-linear effects of age (in years), number of positive nodes, progesterone receptor level (in fmol/ml), oestrogen receptor level (in fmol/ml), and tumor size (in mm), while correcting for tumor grade (grade I vs. grade II and III)
- ▶ Since specific interest is on the **impact of hormone therapy on time to recurrence**, we divide the sample into two groups – main treatment (chemotherapy) and adjuvant treatment (chemotherapy and hormone therapy)
- ▶ **Age, number of positive lymph nodes and progesteron receptor level** are identified as prognostic factors which is in line with the original findings reported by Sauerbrei et al. (1999)
- ▶ Based on the prognostic model, we define **different risk groups**, i.e., low risk (relative risk between 0 and 1), moderate risk (relative risk between 1 and 2) or high risk (relative risk exceeding 2)

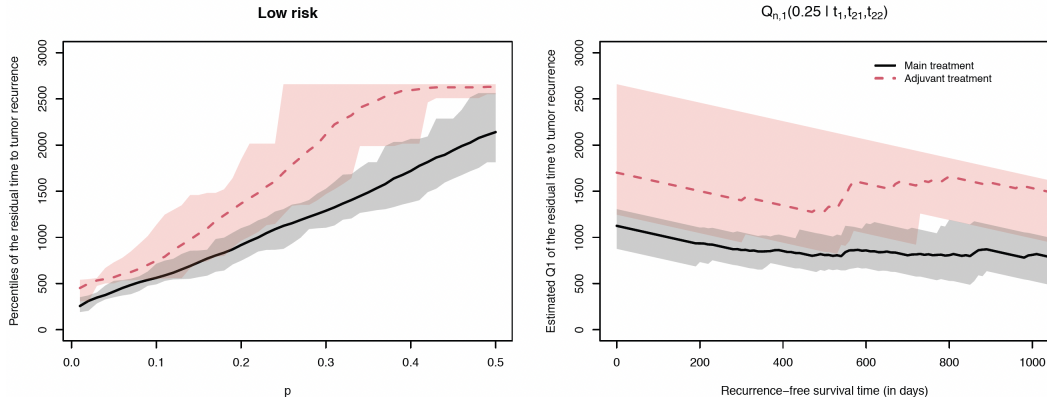


Figure: Estimated quantiles of the time to tumor recurrence ($t_1 = 0$) for individuals in the low risk group and by treatment (main treatment of adjuvant treatment) (left) and estimated first quantile of the residual time to tumor recurrence in relation to the recurrence-free survival time t_1 (right).

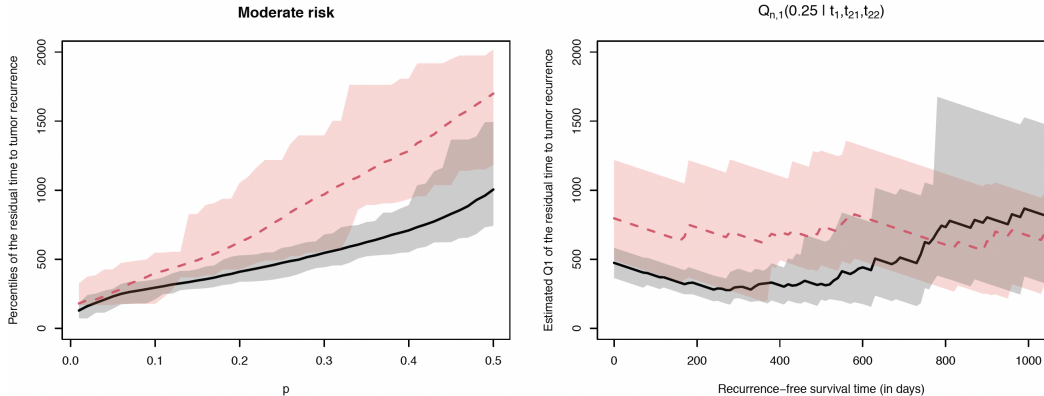


Figure: Estimated quantiles of the time to tumor recurrence ($t_1 = 0$) for individuals in the moderate risk group and by treatment (main treatment of adjuvant treatment) (left) and estimated first quantile of the residual time to tumor recurrence in relation to the recurrence-free survival time t_1 (right).

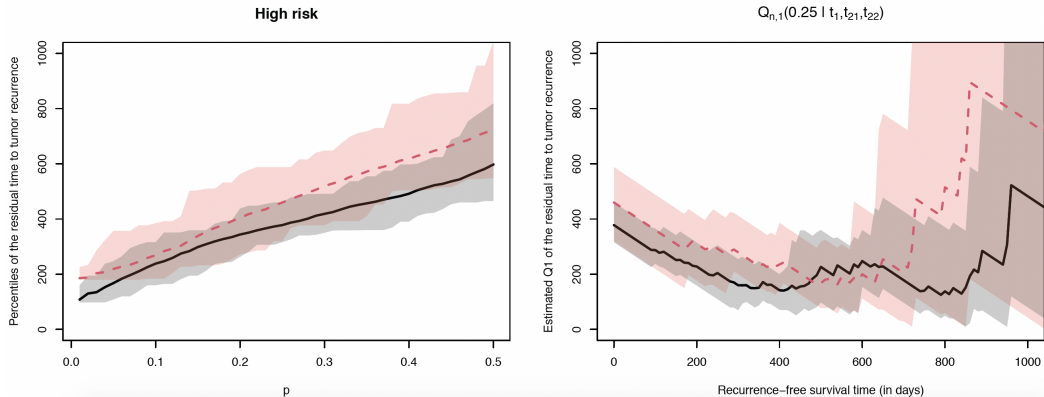


Figure: Estimated quantiles of the time to tumor recurrence ($t_1 = 0$) for individuals in the high risk group and by treatment (main treatment of adjuvant treatment) (left) and estimated first quantile of the residual time to tumor recurrence in relation to the recurrence-free survival time t_1 (right).

- ▶ Estimated percentiles are larger in the adjuvant treatment group (red dashed line) as compared to the main treatment group (black solid line), at least for the low and moderate risk groups, implying **advantage of hormone therapy unless the risk is relatively high**
- ▶ In the **high risk case**, the estimated percentiles of the time to recurrence are very close.
- ▶ In general, the **estimated first quantile is consistently larger in the adjuvant treatment group as compared to the main treatment group** among the different risk groups, **except for individuals in the high risk group**

Discussion

- ▶ Good **finite-sample performance** of proposed estimators
- ▶ Estimators are able to quantify differences in (quantiles of) the conditional residual lifetime distribution across relevant groups, which is of crucial importance in **prognostic research**
- ▶ Patient groups are often constructed based on so-called **prognostic indices**, i.e., clinical classification to discriminate between different prognoses depending on specific prognostic factors
- ▶ Here, we demonstrate that the use of such prognostic indices to construct different risk groups can be used to **understand the association** between T_1 and T_2 .

- ▶ Instead of relying on the **difference of the Kaplan-Meier estimator** for $F_2(t_2)$, evaluated in t_{22} and t_{21} , respectively, to estimate the denominator in equation (1), one could use $\hat{F}(\infty, t_{22}) - \hat{F}(\infty, t_{21})$ as alternative estimator
- ▶ In contrast to the Kaplan-Meier based estimator(s), the estimator itself is **not a well-defined cumulative distribution function** with values approaching one for $t_1 \rightarrow \infty$
- ▶ Although computationally plausible, and providing consistent estimation of the quantiles of the conditional residual lifetime distribution (not shown here), we did not consider such estimators given that their **finite-sample performance was suboptimal** as compared to that of the Kaplan-Meier based estimators

- ▶ Estimators of the quantiles of the conditional residual lifetime distribution can be derived in the **presence of (double) truncation**
- ▶ More specifically, estimation of the conditional distribution function of T_1 , given that $t_{21} < T_2 \leq t_{22}$ could, for example, be based on the **nonparametric maximum likelihood estimator proposed by Efron and Petrosian (1999)**
- ▶ An extension to encompass **random bivariate censoring based on (C_1, C_2)** is an interesting topic for further research
- ▶ Finally, investigating the use of these techniques in the context of **surrogate marker evaluation for survival endpoints** is interesting as well

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